
F ANECDOTAL HALLUCINATION DETECTION EXPERIMENT

As an informal sanity check, we ran a small hallucination detection experiment on Qwen3-8B using Activation Oracles. We prompted the base model with:

“What is the movie ‘Her’? Provide as many specific details and examples as possible (such as names of people, numbers, events, locations, dates, times, etc.).”

The model produced a detailed description that contained three factual errors which we manually identified out of fourteen details.

We then collected activations at individual token positions in the generated answer and passed them to the Activation Oracle with the oracle prompt:

“Are you confident that the statement in this text is accurate?”

In this single example, the Activation Oracle flagged three statements as low confidence, and its judgments aligned with two of the three hallucinated facts.

However, we also found that a simple black-box baseline that asks the underlying model follow-up questions about the accuracy of its own statements performs similarly on this example. Because this setting did not appear to offer a clear advantage for activation-based methods over straightforward prompting, and because our experiment was highly anecdotal, we did not pursue hallucination detection as a primary evaluation task in this work.

G SPARSE AUTOENCODER DATASET EXPERIMENTS

We ran a set of preliminary experiments that used sparse autoencoder (SAE) features as additional training signals for the Activation Oracle. For Qwen3-8B, we trained BatchTopK SAEs (Bussmann et al., 2024) on three layers at 25%, 50%, and 75% depth. Each SAE had 65,000 features and an average L_0 of roughly 80 active features per token. In all cases, we used single-token SAE feature vectors as inputs to the Activation Oracle.

SAE feature explanations. For each SAE, we considered the first 20,000 features and generated automatic natural language explanations using an AutoInterp Detection Evaluation (Paulo et al., 2025) with GPT-5 mini as the judge. We scored each feature with a detection score and retained only features with a score above 80%. This filtering produced approximately 15,000 features per layer, or about 45,000 high quality SAE feature explanations in total. Generating these explanations cost roughly \$1,000 in OpenAI credits, which was one of the practical reasons we did not scale this line of work further.

Binary questions about SAE features. Using each retained SAE explanation, we generated four yes-no questions that capture different aspects of the described concept. For example, if a feature was described as activating on sports stadium scenes, we might ask: “Is this feature related to sports?”, “Is this feature related to stadiums?”, “Is this feature related to libraries?”, and “Is this feature related to hospitals?”. The Activation Oracle received the SAE feature vector as a single-token activation input and was trained to answer these questions. This produced roughly $45,000 \times 4 = 180,000$ supervised training examples.

Max-activating sequence prediction. As a separate objective, we sampled 60,000 features from each SAE and constructed a self-supervised dataset where the AO was trained to predict the five most strongly activating sequences for each feature. Again, the input was a single SAE feature vector and the target was a list of the corresponding maximally activating contexts.

Results. Training on these SAE-derived datasets produced mixed results. We observed modest gains on some held-out classification tasks, but minimal improvements and in some cases small regressions on the out-of-distribution auditing benchmarks that are the focus of our main results. The AO did become significantly better at verbalizing its own SAE features, but it did not surpass the GPT-5 mini baseline used in the original AutoInterp evaluation. Given the limited downstream benefits, the additional implementation complexity, and the nontrivial compute and API cost, we

decided not to include SAE-based datasets in the final Activation Oracle training mixture or main results.